

MA201A — Lecture 01 Notes & Lab Guide

§12.5 Lines in Space · §12.6 Cylinders & Quadric Surfaces — Stewart 9e · Fall 2026

1 · Recap & Visual Intuition

Cylinders & The Missing-Variable Rule

◆ Definition

A **cylinder** is a surface formed by all lines (**rulings**) parallel to a given line passing through a given plane curve. It is any 2D curve extruded infinitely through space.

★ Key Tip

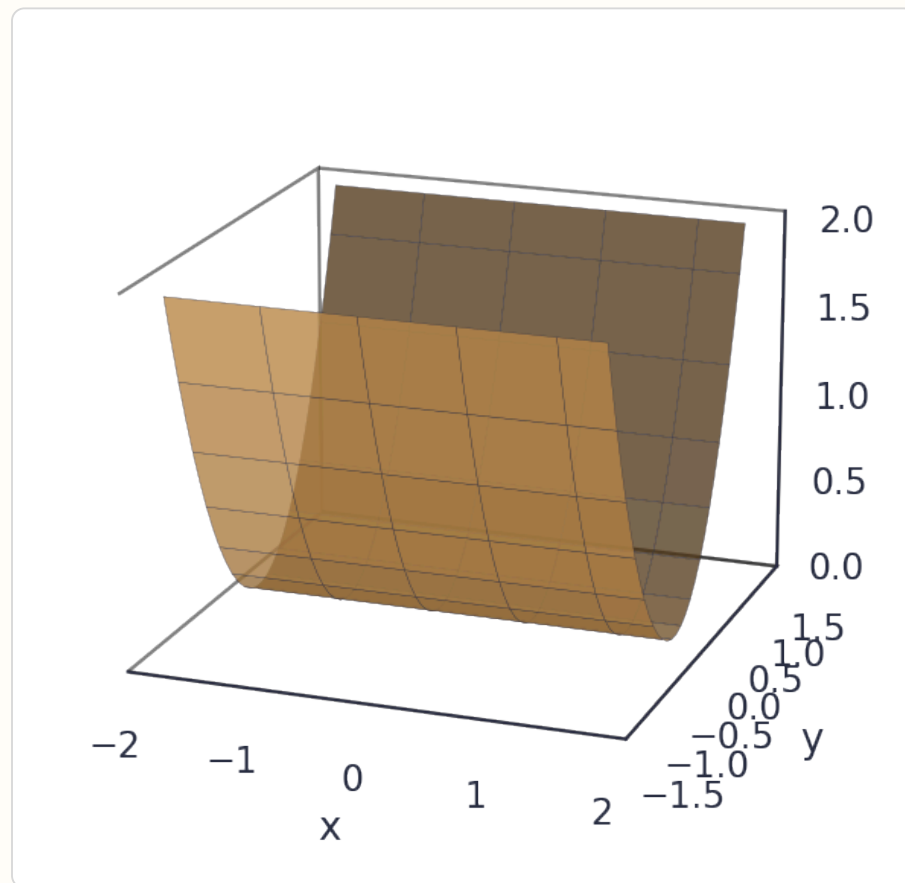
Missing-variable rule — if an equation in $\mathbb{R}^3$ lacks one coordinate, graph the curve in the plane of the two present variables, then extrude it infinitely along the missing axis through all 8 octants.

Circular: $x^2 + y^2 = r^2$ ($\parallel z$)

Elliptic: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a \neq b$)

Parabolic: $z = y^2$ (extruded along x)

Hyperbolic: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (rulings $\parallel z$)



$z = y^2$: parabola in yz -plane extruded along x .

2D Parameterization → 3D Trajectory

Segment from $(0, 1)$ to $(1, 0)$ on line $y = 1 - x$. Setting $x(t) = t \Rightarrow y(t) = 1 - t$:

$$\vec{r}(t) = \langle t, 1 - t \rangle, \quad t \in [0, 1]$$

t	(x, y)	$\vec{r}(t)$	Travel	Interpretation
0	$(0, 1)$	$\langle 0, 1 \rangle$	Start	Base point P_0
$\frac{1}{2}$	$(\frac{1}{2}, \frac{1}{2})$	$\langle \frac{1}{2}, \frac{1}{2} \rangle$	Midpoint	Constant speed
1	$(1, 0)$	$\langle 1, 0 \rangle$	End	Terminal point P_1

★ Parameterization fixes Order, Direction & Speed

The **set** of points is merely a line segment — but parameterization specifies:

1. **Start point** ($t = 0$).
2. **Direction of travel** (towards P_1).
3. **Velocity** ($\vec{v} = \langle 1, -1 \rangle$, speed $|\vec{v}| = \sqrt{2}$).

For the **infinite line**, let $t \in \mathbb{R}$.

Lines in $\mathbb{R}^3$ (Vector, Parametric & Symmetric Forms)

A line L in space is determined by a base point $P_0(x_0, y_0, z_0)$ and direction vector $\vec{v} = \langle a, b, c \rangle \parallel L$.

◆ Vector Form

$$\begin{aligned} \vec{r}(t) &= \vec{r}_0 + t\vec{v} \\ &= \langle x_0 + at, y_0 + bt, z_0 + ct \rangle \end{aligned}$$

◆ Parametric Form

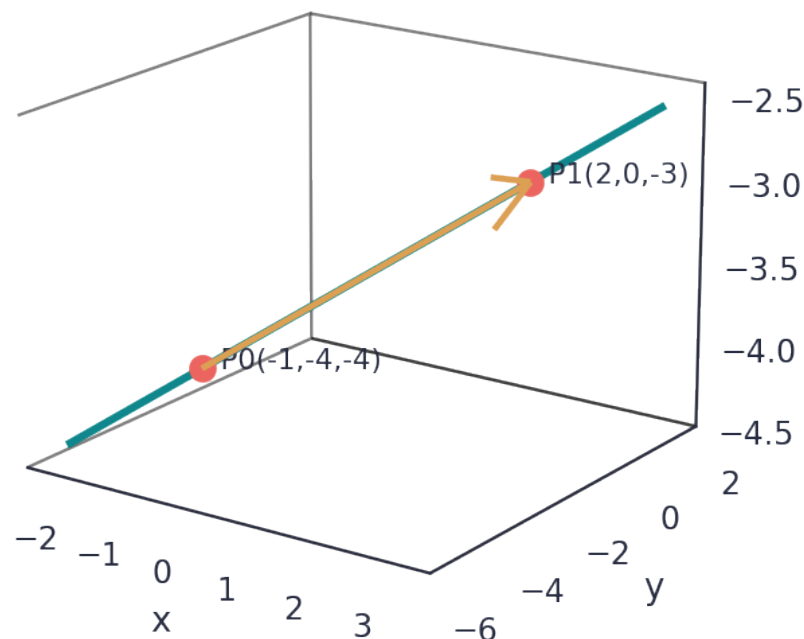
$$\begin{aligned} x &= x_0 + at \\ y &= y_0 + bt, \quad z = z_0 + ct \end{aligned}$$

◆ Symmetric Form ($a, b, c \neq 0$)

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

◆ Segment $P_0 \rightarrow P_1$

$$\begin{aligned} \vec{r}(t) &= (1 - t)\vec{r}_0 + t\vec{r}_1 \\ 0 &\leq t \leq 1 \end{aligned}$$



Line through $P_0(-1, -4, -4)$ and $P_1(2, 0, -3)$, $\vec{v} = \langle 3, 4, 1 \rangle$.

2 · Classification Cheat-Sheet

★ Key Tip

General Second-Degree Equation: $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$.

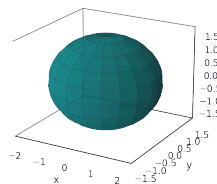
Speed Decision Rule: Count minus signs, check RHS (= 1 vs = 0), and check for an isolated linear variable.

Algebraic Signature	Surface	Axis / Opening	Horizontal Traces	Vertical Traces
All +, = 1	Ellipsoid	Bounded (no opening axis)	Ellipses	Ellipses
Exactly one -, = 1	Hyperboloid, 1 sheet	Along negative variable	Ellipses	Hyperbolas
Exactly two -, = 1	Hyperboloid, 2 sheets	Along positive variable	Ellipses ($ z \geq c$)	Hyperbolas
All degree 2, = 0	Elliptic Cone	Along isolated variable	Ellipses	Hyperbolas / lines
One linear , two quad same sign	Elliptic Paraboloid	Opens along linear variable	Ellipses	Parabolas
One linear , two quad opposite sign	Hyperbolic Paraboloid	Saddle at vertex	Hyperbolas	Parabolas
One variable missing	Cylinder	Rulings missing axis	2D base curve	Parallel lines

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

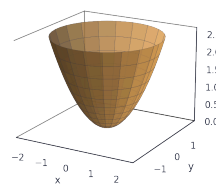
Bounded. All traces ellipses.
 $a = b = c \Rightarrow$ sphere.



Elliptic Paraboloid

$$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

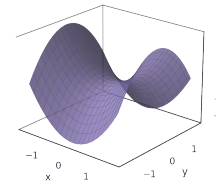
Bowl shape. Up if $c > 0$, down if $c < 0$.



Hyperbolic Paraboloid

$$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

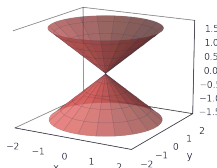
Saddle. Min along one axis, max along other.



Elliptic Cone

$$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

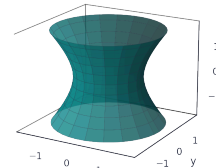
Two lobes apex-to-apex. RHS = 0 giveaway.



Hyperboloid – 1 Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

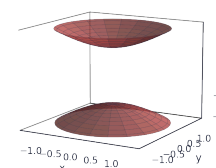
Connected hourglass. **One** minus.



Hyperboloid – 2 Sheets

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Two separate lobes. **Two** minuses.

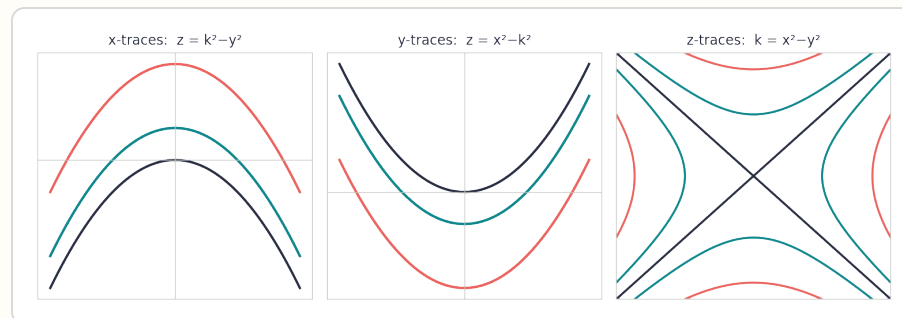


Trace Synthesis: Building $z = x^2 - y^2$

🔗 Anatomy of a Saddle

- **Slice** $x = k$ (planes $\parallel yz$): $z = k^2 - y^2$ (downward parabolas, crest at k^2).
- **Slice** $y = k$ (planes $\parallel xz$): $z = x^2 - k^2$ (upward parabolas, trough at $-k^2$).
- **Slice** $z = k$ (horizontal planes): $k = x^2 - y^2$ (hyperbolas; degenerate crossing lines at $k = 0$).

Opposite curvatures at origin create a **saddle point** (neither local max nor local min).



Slicing $z = x^2 - y^2$: upward along x , downward along y .

3 · Assessment Traps & Lab Reference

⚠️ iClicker Trap: Dimension Ambiguity

$x^2 + y^2 = 1$ is a **circle** in $\mathbb{R}^2$, but a **circular cylinder** in $\mathbb{R}^3$.
 $y = x^2$ is a parabola in $\mathbb{R}^2$, but a **parabolic cylinder** in $\mathbb{R}^3$.
 Always check the ambient space before answering!

⚠️ Exam Trap: Zero Direction Component

If $\vec{v} = \langle a, 0, c \rangle$, the symmetric fraction $\frac{y-y_0}{0}$ is **illegal**.
 Correct form: $\frac{x-x_0}{a} = \frac{z-z_0}{c}$, $y = y_0$.
 Geometrically: the line lies entirely inside the plane $y = y_0$.

⚠️ RHS Sign Governs Everything

$x^2 + y^2 - z^2 = k$:

- $k = 0 \rightarrow$ **Cone** (apex at origin).
- $k = 1 \rightarrow$ **Hyperboloid of 1 sheet** (connected).
- $k = -1 \rightarrow$ **Hyperboloid of 2 sheets** (disconnected).

✓ Assignment 1 Rubric Checklist (Wed Sept 23)

- Write $\vec{v} = \overrightarrow{P_0 P_1}$ explicitly before equations.
- Always declare parameter domain ($t \in \mathbb{R}$ vs $0 \leq t \leq 1$).
- Show complete-the-square steps; centre/vertex earns own mark.
- Verify endpoints $t = 0$ and $t = 1$.

⌘ Maple Lab 1 Reference – Thurs Sept 24 (LH3068)

Essential 3D Plot Commands:

```
with(plots):
# 1. 3D Spacecurve (parametric line segment):
spacecurve([-1+3*t, -4+4*t, -4+t], t=0..1, thickness=3, color=red);

# 2. Implicit Quadric Surface:
implicitplot3d(x^2/4 + y^2/9 + z^2 = 1, x=-3..3, y=-4..4, z=-2..2);

# 3. Explicit Surface:
plot3d(x^2 - y^2, x=-2..2, y=-2..2, style=patchcontour);
```

Intersection Curves & Constrained Scaling:

```
# Curve of intersection: z = x^2+y^2 and z = 4
spacecurve([2*cos(t), 2*sin(t), 4], t=0..2*Pi, thickness=3);

# Display multiple objects together:
display([surface, curve], scaling=constrained);
```

Golden Lab Habit: Always append `scaling=constrained` so equal distances along x, y, z appear equal on screen (prevents spheres from looking like footballs).